1) Server side code for calculator(addition and subtraction)

```
package calculator;


import javax.jws.WebService;

import javax.jws.WebMethod;


@WebService

public class CalculatorService {


    @WebMethod

    public int add(int a, int b) {

        return a + b;

    }


    @WebMethod

    public int subtract(int a, int b) {

        return a - b;

    }

}
```

Client side code(java class)

```
package calculatorclient;


import calculator.CalculatorService;
import calculator.CalculatorService_Service;


public class CalculatorClient {


    public static void main(String[] args) {


        CalculatorService_Service service = new CalculatorService_Service();
        CalculatorService port = service.getCalculatorServicePort();


        int a = 20;
        int b = 10;


        int sum = port.add(a, b);
        int sub = port.subtract(a, b);


        System.out.println("Addition = " + sum);
        System.out.println("Subtraction = " + sub);
    }
}
```

2) server side code for calculator(multiplication and division)

```
package calculatorclient;


import calculator.CalculatorService;
import calculator.CalculatorService_Service;


public class CalculatorClient {


    public static void main(String[] args) {


        CalculatorService_Service service = new CalculatorService_Service();
        CalculatorService port = service.getCalculatorServicePort();


        int a = 20;
        int b = 4;


        int mul = port.multiply(a, b);
        double div = port.divide(a, b);


        System.out.println("Multiplication = " + mul);
        System.out.println("Division = " + div);
    }
}
```

Code for client side

```
package calculatorclient;


import calculator.CalculatorService;
import calculator.CalculatorService_Service;


public class CalculatorClient {


    public static void main(String[] args) {


        CalculatorService_Service service = new CalculatorService_Service();
        CalculatorService port = service.getCalculatorServicePort();


        int a = 20;
        int b = 4;


        int mul = port.multiply(a, b);
        double div = port.divide(a, b);


        System.out.println("Multiplication = " + mul);
        System.out.println("Division = " + div);
    }
}
```

3) server side code for square number

```
package square;


import javax.jws.WebService;

import javax.jws.WebMethod;


@WebService(serviceName = "SquareWS")

public class SquareWS {


    @WebMethod

    public int square(int n) {

        return n * n;

    }

}
```

Client side code for square

```
<%@page import="square.SquareWS"%>

<%@page import="square.SquareWS_Service"%>

<html>

<body>

<h2>Square of a Number (SOAP)</h2>


<form method="post">

Enter Number:

<input type="text" name="num">

<input type="submit" value="Find Square">

</form>


<%

if(request.getParameter("num") != null){

   int n = Integer.parseInt(request.getParameter("num"));


   SquareWS_Service service = new SquareWS_Service();

   SquareWS port = service.getSquareWSPort();


   int result = port.square(n);

%>
```

```
<h3>Square = <%= result %></h3>


<% } %>


</body>

</html>
```

Client code for sqare using java

```
package squareclient;

import java.util.Scanner;

import square.SquareWS;

import square.SquareWS_Service;


public class Main {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        System.out.print("Enter number: ");

        int n = sc.nextInt();

        SquareWS_Service service = new SquareWS_Service();

        SquareWS port = service.getSquareWSPort();

        int result = port.square(n);

        System.out.println("Square = " + result);

    }

}
```

4) Server side code for largest number

```
package largest;


import javax.jws.WebService;
import javax.jws.WebMethod;


@WebService(serviceName = "LargestWS")
public class LargestWS {


    @WebMethod
    public int findLargest(int a, int b) {
        if (a > b)
            return a;
        else
            return b;
    }
}
```

Client side code for largest number

```
package largestclient;


import java.util.Scanner;
import largest.LargestWS;
import largest.LargestWS_Service;
```

```
public class Main {
   public static void main(String[] args) {
      Scanner sc = new Scanner(System.in);
      System.out.print("Enter first number: ");
      int a = sc.nextInt();
      System.out.print("Enter second number: ");
      int b = sc.nextInt();
      LargestWS_Service service = new LargestWS_Service();
      LargestWS port = service.getLargestWSPort();
      int result = port.findLargest(a, b);
      System.out.println("Largest number = " + result);
   }
}
```

Client side code for largest number using jsp

```
<%@page import="largest.LargestWS"%>
<%@page import="largest.LargestWS_Service"%>

<html>
<body>

<h2>Largest of Two Numbers (SOAP)</h2>

<form method="post">
Enter First Number:
```

```
<input type="text" name="n1"><br><br>

Enter Second Number:
<input type="text" name="n2"><br><br>

<input type="submit" value="Find Largest">
</form>

<%
if(request.getParameter("n1") != null && request.getParameter("n2") != null){

   int a = Integer.parseInt(request.getParameter("n1"));
   int b = Integer.parseInt(request.getParameter("n2"));

   LargestWS_Service service = new LargestWS_Service();
   LargestWS port = service.getLargestWSPort();

   int result = port.largest(a, b);
%>

<h3>Largest Number = <%= result %></h3>

<% } %>

</body>
```

```
</html>
```

5) Server side code for cube

```
package cube;

import javax.jws.WebService;
import javax.jws.WebMethod;

@WebService(serviceName = "CubeWS")
public class CubeWS {

    @WebMethod
    public int cube(int n) {
        return n * n * n;
    }
}
```

Client side code for cube

```
<%@page import="cube.CubeWS"%>
<%@page import="cube.CubeWS_Service"%>

<html>
<body>

<h2>Cube of a Number (SOAP)</h2>

<form method="post">
Enter Number:
<input type="text" name="num">
<input type="submit" value="Find Cube">
</form>

<%
if(request.getParameter("num") != null){

   int n = Integer.parseInt(request.getParameter("num"));

   CubeWS_Service service = new CubeWS_Service();
   CubeWS port = service.getCubeWSPort();

   int result = port.cube(n);
```

```
%>

<h3>Cube = <%= result %></h3>

<% } %>

</body>
</html>
```